

IBUKI Circle

One breath. One tap. Your circle responds. A multimodal recovery and prevention platform for people navigating substance use disorders — and the caregivers standing beside them.

The worst time to type

- Typing fails exactly when cognitive load peaks — a craving, a panic attack, a suspected overdose.
- In that moment nobody needs another chatbot. They need one safe next action.
- An emergency cannot wait for a model response — and a helpline number should never depend on model memory.
- Caregivers often know something is wrong, but not what to say.

**“AI personalizes the support.
Deterministic code controls safety.”**

— the one design rule behind every screen

01

One tap, one plan

From distress to a connected plan

One tap in

A tap, a spoken sentence, or a short message enters one validated endpoint — /api/intervene. There is no prompt to compose.

Risk checked first

A deterministic safety router classifies the situation before any model runs, then hands it to one of four specialists.

A connected plan

One live model call returns coping steps, a paced-breathing guide, an editable message for a trusted person, and verified helplines.

Multimodal, human-centred

Speak, listen, converse

Speech input, read-aloud, and a live speech-to-speech voice that keeps talking while it places breathing guides, support messages, and helplines directly on screen.

Eight languages, two modes

Plans generate live in English, Tamil, Hindi, Bengali, Telugu, Marathi, Kannada, and Malayalam. Caregiver mode returns what to say, what to avoid, and warning signs.

112

A suspected overdose never waits for a model.

Deterministic code shows India's 112 action and reviewed response steps immediately — the model cannot lower the risk level, invent a helpline, or delay the response.

02

Under the hood

One pipeline, four specialists

One intervention API

Every tap, message, and transcript is validated, routed by the deterministic safety router, and assigned to Safety Guardian, Recovery Coach, Caregiver Guide, or Resource Navigator.

Model-agnostic intent

GPT-5.6 Terra or Gemini — behind one provider boundary with automatic cross-provider fallback — returns strict structured intent, validated by JSON Schema and Zod.

Code owns actions

A deterministic compiler renders only six approved widget types. The model never controls React components, calls, messages, location, or emergency routing.

The stack

	LAYER	TECHNOLOGY	ROLE
01	Application	Next.js 16 App Router · React 19	UI, server routes, streaming
02	GenAI	GPT-5.6 Terra · Gemini 3.6 Flash · Realtime · TTS	Structured intent, live voice, read-aloud
03	Safety	TypeScript router · JSON Schema · Zod 4	Deterministic routing, strict contract
04	Tooling	Nx 23 · pnpm · Tailwind 4 · Vitest 4	Cached CI, 65 tests, Vercel + Docker

How we built it

Double validation

Strict JSON Schema at the API layer, Zod re-validation, then per-agent widget and resource allow-lists — invalid output is rejected, never substituted. 15 s hard timeouts; failover timing owned end-to-end.

CI quality gates

Frozen install → zero-warning lint → 65 tests across 14 files → production dependency audit → build → full-history secret scan. Emergency-phrases sweeps, schema contracts, and the rate limiter all covered.

Deploy-early discipline

13 production deploys in one day, every feature verified live before it was claimed. Localization shipped safely: 45 UI keys × 8 languages as tested constants; medical protocols stay English by design.

Contained voice, private core

Live voice, contained

OpenAI Realtime over WebRTC on a 10-minute ephemeral token — the server key never leaves. Five allow-listed screen tools, argument-validated and clamped; the emergency tool re-enters the same safety router.

Private by design

No database: transcripts, audio, location, contacts, and recovery history are never persisted. Server-side secrets, bounded input, no-store responses, CI dependency audit and secret scanning.

Stateless today, ready to scale

Stateless core

No database and no sessions to migrate — the standalone Next.js build runs identically on Vercel serverless and the node:22-alpine Docker image, so compute scales horizontally.

Efficient under load

One structured model call per non-emergency, low reasoning effort, zero model latency on Level 1. Model-backed routes carry per-instance rate limits and 16 KB bounded requests.

The growth path

Distributed quotas on shared storage, multi-region deployment, country-specific registries, reviewed translations — and the provider boundary routing on availability, cost, and latency.

03

What's next

Future scope, in priority order

01 · My Circle loop

A caregiver opens a short-lived link and taps “I’m on my way” — reflected live on the person’s screen. Upstash Redis, 24 h TTL, no PII server-side.

02 · Onboarding & persona

A skippable three-step profile — coping preferences, trusted-contact label, language — stored locally; the orchestrator already consumes every field.

03 · Peer support forum

A moderated, anonymous community space with real persistence — and safety-router screening on every post before it lands.

04 · Clinical legitimacy

Professionally reviewed protocols, clinically validated Tamil and Hindi emergency translations, and a standing clinician advisory loop.

05 · Recovery journey

Non-punitive daily check-ins, a personal safety card, and engagement-based progress — never shame-based streaks.

06 · Production hardening

Distributed rate limiting, CSP, signed sessions, a deployed E2E and axe accessibility suite, and a PWA offline crisis mode.

Thank you

IBUKI Circle reduces the time between a moment of distress and a safe next action. Try it live at web-delta-three-92.vercel.app — demo [ibuki-demo / circle2026](#), or continue as guest.